Project Design Phase
Solution Architecture

Date	27 February 2026
Team ID	LTVIP2026TMIDS41352
Project Name	ShopEZ:One-Stop Shop for Online Purchases
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture of ShopEZ bridges the gap between user shopping challenges and a scalable, secure technology solution. It defines how frontend, backend, database, and external services interact to deliver a seamless e-commerce experience.

ShopEZ follows a three-tier architecture consisting of:

1. **Presentation Layer**
2. **Application Layer**
3. **Data Layer**

This structure ensures modularity, scalability, maintainability, and secure communication.

Example - Solution Architecture Diagram:

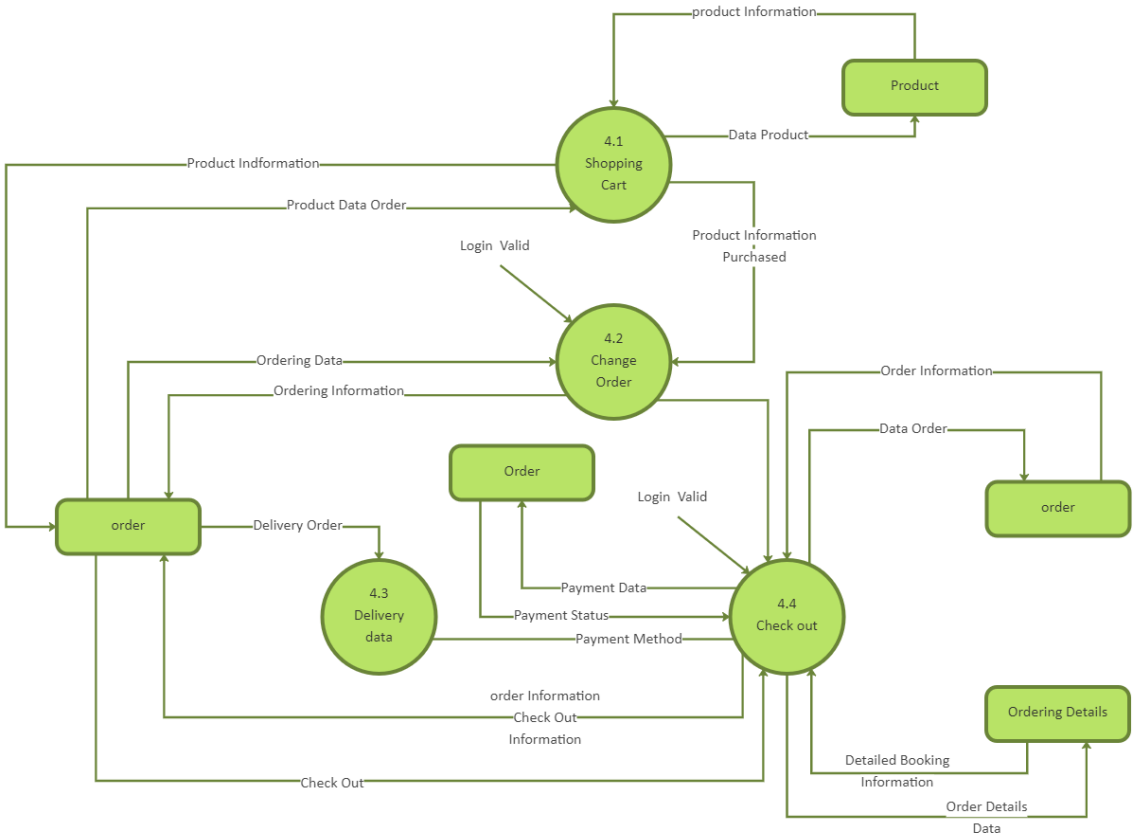

Figure 1: Architecture and data flow of the voice patient diary sample application

1. **Reference:** AWS Industries Blog – Voice Applications in Clinical Research Powered by AI on AWS (Architecture and Design Considerations).
   Available at: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/
   Referenced for architecture design considerations and example architecture diagrams.
2. C4 Model for Software Architecture.
   Available at: https://c4model.com/
   Used to understand layered architectural diagrams and structured representation of system components.
3. AWS Architecture Center.
   Available at: https://aws.amazon.com/architecture
   Used for best practices in cloud-based architecture design, including scalability and security.
4. Medium – How to Draw Useful Technical Architecture Diagrams.
   Available at: https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d
   Referenced for guidelines on creating simple and effective architecture visuals.
5. MERN Stack Official Docs.
   • React: https://react.dev/
   • Node.js: https://nodejs.org/en/docs/
   • Express: https://expressjs.com/
   • MongoDB: https://www.mongodb.com/docs/
   Referenced for component-specific implementation standards and architecture.
6. JWT.io Documentation (JSON Web Tokens).
   Available at: https://jwt.io/
   Used as reference for secure authentication mechanisms in the proposed solution.
7. MongoDB Atlas Documentation.
   Available at: https://www.mongodb.com/cloud/atlas
   Referenced for cloud database deployment guidelines and architectural considerations.